**DOP : 17/08/26 / B3 , 18/08/26/B2 , 19/08/26/B1 , 20/08/26 / B4**
**DOS :**

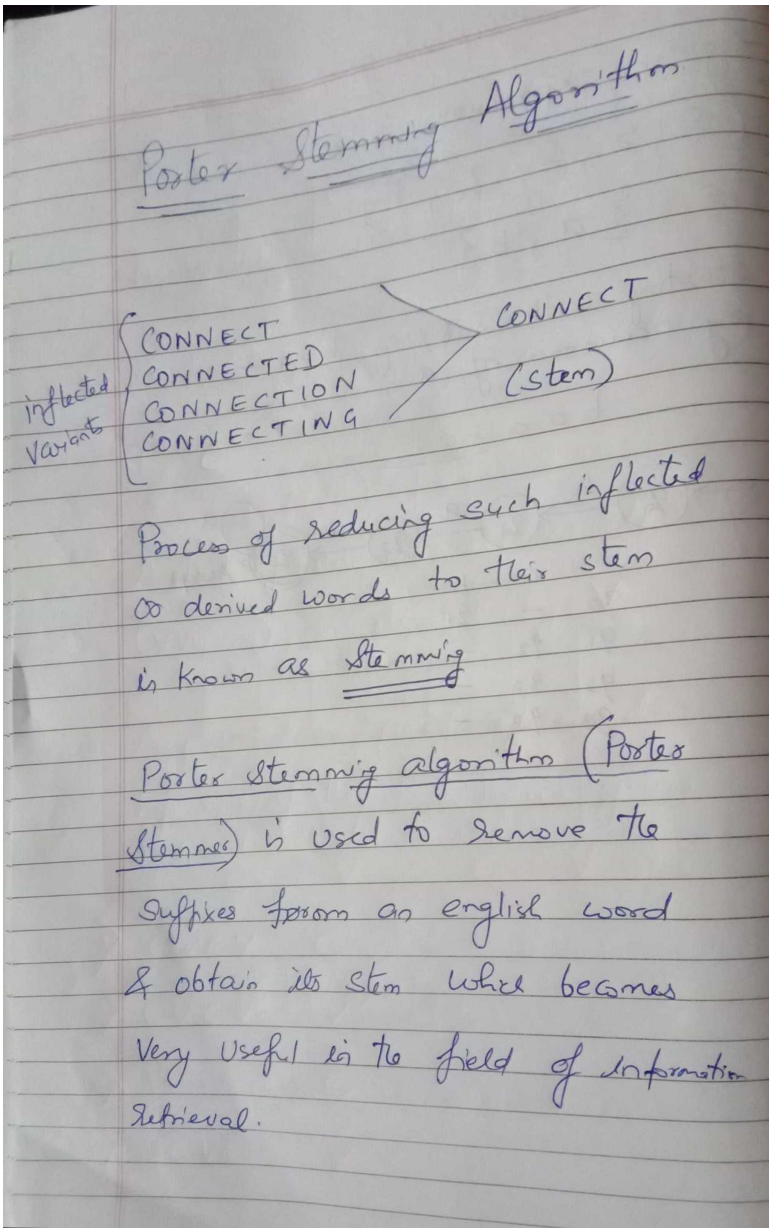
**NLP PRACTICAL MANUAL - Prof. D V Chandran**

**EXPT 5 : Implementing Stemming using Porter Stemming Algorithm / Lancaster / Snowball with NLTK (English) & iNLTK & Stanford (Regional Language) in Python**

**AIM :**
**5.A] WAP to implement stemming using Porter stemming algorithm / Lancaster / Snowball with NLTK (English) .**

**5.B] WAP to implement stemming using Porter stemming algorithm / Lancaster / Snowball with iNLTK & Stanford (Regional Language) in Python**

**THEORY : Refer to the Porter Stemming Algorithm shared on Google classroom.**



**PORTER STEMMING ALGORITHM :**

## The Algorithm

### Step 1a

| | | | |
|---|---|---|---|
| 1. | SSES | → | SS |
| 2. | IES | → | I |
| 3. | SS | → | SS |
| 4. | S | → | |

### Step 1b

| | | | |
|---|---|---|---|
| 1. | (m>0) EED | → | EE |
| 2. | (*v*) ED | → | |
| 3. | (*v*) ING | → | |

If the second or third of the rules in Step 1b is successful, the following is performed.

| | | | |
|---|---|---|---|
| 1. | AT | → | ATE |
| 2. | BL | → | BLE |
| 3. | IZ | → | IZE |
| 4. | (*d and not (*L or *S or *Z)) | → | single letter |
| 5. | (m=1 and *o) | → | E |

### Step 1c

| | | | |
|---|---|---|---|
| 1. | (*v*) Y | → | I |

## Step 2

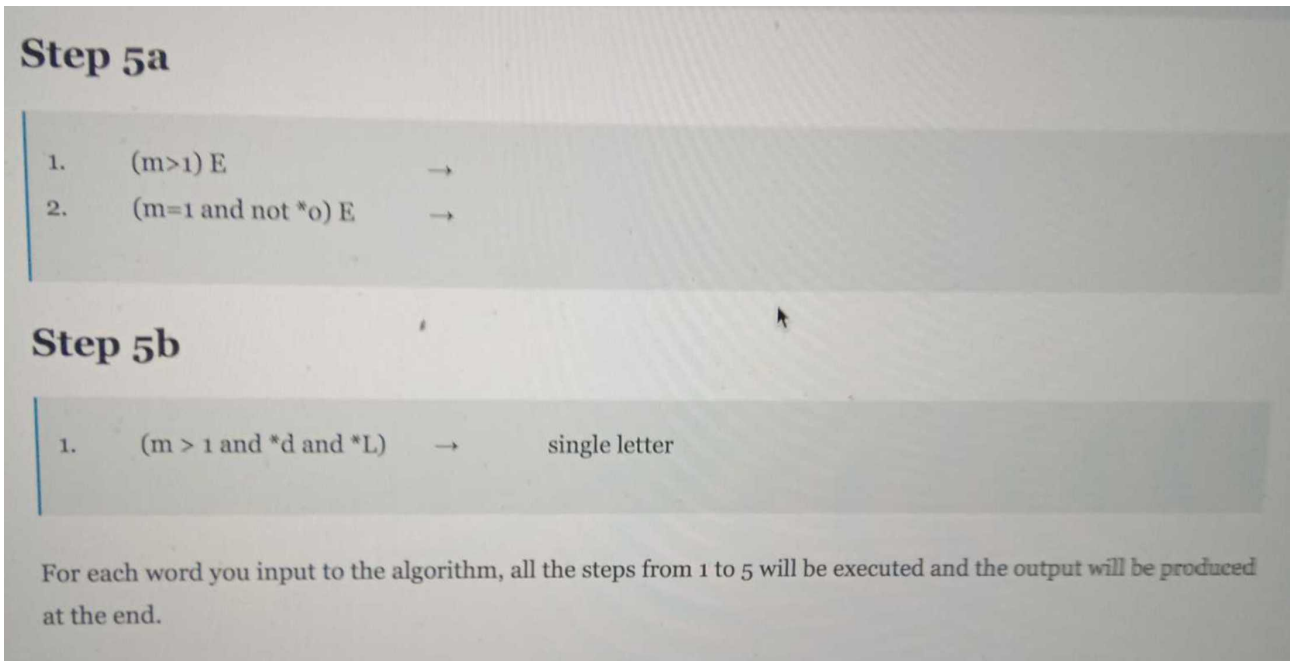
| | | | |
|---|---|---|---|
| 1. | (m>0) ATIONAL | → | ATE |
| 2. | (m>0) TIONAL | → | TION |
| 3. | (m>0) ENCI | → | ENCE |
| 4. | (m>0) ANCI | → | ANCE |
| 5. | (m>0) IZER | → | IZE |
| 6. | (m>0) ABLI | → | ABLE |
| 7. | (m>0) ALLI | → | AL |
| 8. | (m>0) ENTLI | → | ENT |
| 9. | (m>0) ELI | → | E |
| 10. | (m>0) OUSLI | → | OUS |
| 11. | (m>0) IZATION | → | IZE |
| 12. | (m>0) ATION | → | ATE |
| 13. | (m>0) ATOR | → | ATE |
| 14. | (m>0) ALISM | → | AL |
| 15. | (m>0) IVENESS | → | IVE |
| 16. | (m>0) FULNESS | → | FUL |
| 17. | (m>0) OUSNESS | → | OUS |
| 18. | (m>0) ALITI | → | AL |
| 19. | (m>0) IVITI | → | IVE |
| 20. | (m>0) BILITI | → | BLE |

## Step 3

| | | | |
|---|---|---|---|
| 1. | (m>0) ICATE | → | IC |
| 2. | (m>0) ATIVE | → | |
| 3. | (m>0) ALIZE | → | AL |
| 4. | (m>0) ICITI | → | IC |
| 5. | (m>0) ICAL | → | IC |
| 6. | (m>0) FUL | → | |
| 7. | (m>0) NESS | → | |

# Step 4

1. (m>1) AL →
2. (m>1) ANCE →
3. (m>1) ENCE →
4. (m>1) ER →
5. (m>1) IC →
6. (m>1) ABLE →
7. (m>1) IBLE →
8. (m>1) ANT →
9. (m>1) EMENT →
10. (m>1) MENT →
11. (m>1) ENT →
12. (m>1 and (*S or *T)) ION →
13. (m>1) OU →
14. (m>1) ISM →
15. (m>1) ATE →
16. (m>1) ITI →
17. (m>1) OUS →
18. (m>1) IVE →
19. (m>1) IZE →




**ALGORITHM :**

**INPUT - OUTPUT : (with all 3 algorithm in English language / Regional language)**

**INPUT : HUMANITARIAN**
**OUTPUT : HUMAN**

**INPUT : RESEMBLANCE**
**OUTPUT : RESEMBLE**

**CONCLUSION : Conclude with the findings / observation in use of three - Porter stemming Algorithm / Lancaster / Snowball**

**PROGRAM AND OUTPUT PRINTOUT**